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Hybrid argon plasma coagulation after endoscopic mucosal resection—some caveats in the comparison with snare tip soft coagulation



We read with great interest the study by Motchum et al,¹ who analyzed the performance of hybrid argon-plasma coagulation (APC), a technique whereby APC is preceded by submucosal injection of saline through a high-pressure water jet² after EMR of large colonic lesions. The reduction of post-EMR recurrence rates through the use of APC or snare-tip soft coagulation (STSC) on the normal-appearing scar margins is well documented.^{3,4} Prior studies had also suggested that STSC may be preferable to APC.⁵

In this study, Motchum and et al¹ found lower recurrence in the first surveillance colonoscopy after hybrid APC (2.2%) than that reported for STSC (6.0%, in a recent analysis), as well as a lower percentage of serious side effects (3.6% versus 8.5%).⁶ These results are excellent and allow hypothesizing that hybrid APC may be the method of choice, potentially because of the safer ablation of deeper tissue.⁷ Notwithstanding, some caveats arise, including the small sample size, the single-arm design, and the lack of standardization of the hybrid APC technique, given that most patients (72.0%) underwent ablation of both scar surface and margins, hampering comparison with prior STSC studies. No recurrence at 6 months was observed in another recent pilot study⁸ of hybrid APC for colonic large lesions where scar surface ablation was performed in all patients, being plausible that such ablation may justify the lower recurrence rate. Another drawback is the lack of cost effectiveness analysis. In fact, although STSC does not involve additional costs, hybrid APC carries an incremental cost effectiveness ratio of \$158.7/1.0% (or \$603.0 per procedure for a 3.8% decrease in recurrence), and fixed costs of around \$50,000 for Erbejet2 and APC3 modules (prices for Portugal). Additional studies are needed to compare head-to-head standardized STSC, APC, and hybrid APC, to evaluate characteristics of the lesions that may lead to selecting 1 technique over another, and to evaluate cost effectiveness.

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Are sex-specific benchmarks for clinically significant serrated polyp detection warranted?



To the Editor:

We read the recently published article by Anderson et al¹ and the accompanying editorial by Macaron et al² with interest. The study demonstrated that higher detection rates of clinically significant serrated polyps (CSSP-DRs) are associated with lower rates of postcolonoscopy colorectal cancer. The findings support using CSSP-DR as an independent quality measure for colonoscopy. Although target adenoma detection rates are largely agreed upon, there remains heterogeneity of serrated polyp definitions in various

populations. Potential influences of ethnicity, age, sex, patient adiposity, colonoscope lens and processor quality, quality of bowel preparation, and polyp location are commonly not considered in discussions of optimal benchmarks for CSSP-DR. We hereby share the findings from our recently published study,³ which raise interesting questions for consideration in future studies.

In our study, we observed significant sex disparities in CSSP-DRs in a general outpatient colonoscopy population ($n = 2097$). When male and female individuals were compared, CSSP detection was 15% and 20%, respectively ($P < .05$), and sessile serrated lesions (SSLs) were 11% and 17%, respectively ($P < .05$). CSSPs and SSLs were independently associated with female sex and synchronous adenomas and SSLs also with body mass index $\geq 25 \text{ kg/m}^2$. SSL and adenoma detection were highly correlated ($r = 0.83$, $P = .002$). Therefore, whether CRC develops through a serrated or an adenoma pathway, CSSP-DR may provide a window to overall quality of colonoscopy. The question we pose is whether male and female individuals have inherently different susceptibilities to CSSPs. Studies of screening populations aged ≥ 50 years or with positive fecal immunologic test results have a bias toward greater proportions of males with adenomas.⁴ Furthermore, the fecal immunologic test is not particularly useful for CSSP detection.⁵ There hence remains a scarcity of research regarding patient and procedure characteristics associated with CSSP detection. Inasmuch as the diagnostic criteria for serrated polyps have recently been revised,⁶ future studies should examine these characteristics in more detail. It is possible that artificial intelligence in colonoscopy may have a role in defining CSSP-DR, including in populations non-screening colonoscopy.

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Response:



We thank Ismail et al¹ for their comments and for highlighting the importance of identifying risk factors associated with serrated polyps. Inasmuch as sessile serrated polyps can be difficult to distinguish from hyperplastic polyps,² we have proposed clinically significant serrated polyps (CSSPs) as a surrogate outcome for sessile serrated polyps (SSPs), which include any sessile serrated polyps, traditional serrated adenomas, large ($\geq 1 \text{ cm}$) hyperplastic polyps (HPs), or those proximal HPs $> 5 \text{ mm}$.³ As published in a prior study,⁴ an analysis of data from the New Hampshire Colonoscopy Registry population demonstrated that higher body mass index and also smoking were strongly associated with CSSP.⁵ Although older age and male sex have been observed to be associated with conventional adenomas, these factors do not appear to be major factors associated with serrated polyps. It is possible that some confounding factors such as smoking, which has been strongly linked with serrated histologic features^{5,6} and with BRAF mutations and the CpG island methylator phenotype,⁷ may explain the differences observed for the prevalence of CSSPs in male and female patients in the authors' analyses. We agree with the authors that identifying risk factors would help in stratifying patients for screening. Potentially, those at higher risk for serrated or conventional adenomas could be screened with colonoscopy, whereas other low-risk patients could receive noninvasive testing. Finally, we also agree with the authors that attention to clinically significant serrated